SHASANKA SHEKHAR PADHI

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Senior Specialist

Summary

A Bioinformatics professional with 3+ years of experience in AI/ML model development, biomarker identification, and knowledge graph applications. Skilled in **GenAI, LLM, knowledge graph analytics and core bioinformatics solutions** to drive data driven insights, enhance decision making, and deliver scalable solutions through cross functional collaboration across diverse domains. Focused on advancing computational research that integrates AI and data-driven science to solve complex problems across biology, healthcare, and beyond.

Technical skills

Programming languages: Python | R | Cypher | Shell Scripting
AI/ML- Frameworks: PyTorch | Scikit-learn | TensorFlow | SciPy | LLM | GenAI | GraphRAG
Deployment: Docker | GIT | RESTAPI | LangChain
Databases: Neo4J | MongoDB | ChromaDB
Multiomics: Transcriptomics (RNA Seq and scRNA Seq) | Genomics (WGS)

Work Experiences

Senior Specialist — Jun '23 - Present

Syngene — Bengaluru

Received SPOT award for advancing bioinformatics workflow.

- **AI driven antibody sequence generation**: Developed an **AI** driven pipeline for antibody sequence generation, integrated with sequence optimization and structural validation tools to design high affinity, **developable antibodies** with therapeutic potential.
- **Knowledge Graph**: Built a KG with open source biological/RWE/clinical data, harmonized with **controlled vocabularies** for each entity. Application included drug repurposing, target identification, safety assessment for toxicity and organ wise stratification, reducing months of work to weeks.
- **Machine Learning models**: Developed an **automated ML** pipeline to build Quantitative Structure-Property Relationship (**QSPR**) model for drug property prediction, that helped reducing dependency on data scientist for model building and increased capabilities across departments.
- **GWAS - AllOfUs cohort**: Developed a Cromwell (WDL) pipeline to scale up association studies across 3000+ traits using **EHRs** in the cohort to identify **statistically significant** genetic biomarkers.
- **Machine Learning driven biomarker identification:** Developed a computational pipeline for automated biomarker identification using **TCGA** database, by training classical **ML** models using **RNA seq** data, to identify diagnostic and prognostic signatures in BRCA, LIHC, LUAD and PRAD. Enabling faster patient stratification for precision medicine applications.
- **Network analysis:** High throughput drug combination (synergy) assessment using network proximity analysis.
- ***In-silico* KO/perturbation**: Developed a high throughput **Boolean model simulation** pipeline for in silico gene knockout/perturbation experiments, using RNA seq data to initialize the system states supporting data driven therapeutics which enhances precision in target prioritization.
- **Structure-based druggability**: Developed a structure based druggability prediction pipeline leveraging parallel processing to accelerate searches across a database of known **binding pockets**, enabling rapid identification of similar sites to assess protein druggability.
- **Spectral deconvolution and compound detection:** Developed an automated compound identification pipeline leveraging **spectral data**, matched against a processed in-house reference compound database.

Project Associate — Oct '22 - Jun '23

Centre for Brain Research, IISc — Bengaluru

Performed quality control, variant association and cross population studies using **GenomeIndia** GWAS data.

Single Cell Curation Intern — Aug '22 - Oct '22

Elucidata — Remote

Curation and standardizing annotations for **scRNA seq** data for large scale model development.

Personal projects

- **CAMDA challenge:** Constructed a **Temporal Knowledge Graph** from diabetes patient records (**EHR**) using **Neo4j**, integrated with llama3 to setup a **GraphRAG** workflow for **AI driven medical applications**.
- **GenAI & RAG**: Developed a biomedical research assistant that streamlines literature exploration using **llama3** with a **chatbot** for natural language Q&A
- **NLP**: Autism Spectrum Disorder (ASD) classification from caregiver-written behavioural text
- **NeurIPS_Open Polymer Prediction Challenge:** QSPR models for polymer property prediction.
- Built a **Deep learning** model (with **PyTorch**) using protein sequence embeddings for protein classification

Education

Master of Technology - Bioinformatics

University of Hyderabad — Hyderabad

9.05 CGPA - GATE 2022 (AIR : 309)

Bachelor of Engineering - Biotechnology

Birla Institute of Technology — Ranchi

7.44 CGPA - GATE 2020 (AIR : 767)

Certifications

- Big data, genes and medicine - The state university of New York (Coursera)
- Artificial Intelligence (Syngene)